

Lexical in Tiger Language

- Along with the *tiger.lex* file you should turn in a document that describes the followings:
 - How you handle comments
 - How you handle strings
 - Error handling
 - End-of-file handling
 - Other interesting features of your lexer

Lexical in Tiger Language

整理一下tiger语言的基本模式(具体见附录):

- Reserved words
 - while, for, to, break, let, in, end, function, var, type, array, if, then, else, do, of, nil
- Punctuation symbols
 - , : ; () [] { } . + - * / = <> < <= > >= & | :=
- Identifier
 - A sequence of letters, digits, and underscores
 - Starting with a letter

Lexical in Tiger Language

对词法分析器输出能做什么处理.

- Comments
 - May appear between any two tokens
 - Start with /* and end with */
 - May be nested
 - detect unclosed comments (at end of file)
- Integer literal
 - No negative value

词法出错, 该怎么处理.

Lexical in Tiger Language

- String literal
 - A sequence between quotes (") of zero or more printable characters, spaces, or
 - escape sequences
 - \n, \t: linefeed, tab
 - \ddd: ddd is a 3-octal number
 - \", \\
 - \f__f\ ^{转义符}: this sequence is ignored, where f__f stands for sequence of one or more formatted characters including at least space, tab, or newline, form feed

Lexical in Tiger Language

- String literal
 - escape sequences
 - `\^c`:
 - `^@` (0, null, NUL, `\0`), mark the end of a string
 - `^G` (7, bell, BEL, `\a`), emit a warning of some kind
 - `^H` (8, backspace, BS, `\b`), may overprint the previous character
 - `^I` (9, horizontal tab, HT, `\t`), moves the printing position right to the next tab stop
 - `^J` (10, line feed, LF, `\n`), moves the print head down one line
 - `^K` (11, vertical tab, VT, `\v`), vertical tabulation

Lexical in Tiger Language

- String literal
 - escape sequences
 - $\backslash^c$:
 - $\backslash^L$ (12, form feed, FF, $\backslash f$), to cause a printer to eject paper to the top of the next page, or a video terminal to clear the screen.
 - $\backslash^M$ (13, carriage return, CR, $\backslash r$), moves the printing position to the start of the line, allowing overprinting.
 - $\backslash^Z$ (26, Control-Z, SUB, EOF). Acts as an end-of-file for the Windows text-mode file i/o.
 - $\backslash^$ (27, escape, ESC, $\backslash e$ (GCC only)). Introduces an escape sequence
 - Detect unclosed string

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检测是否处理.